

Day 05 — Ethernet and MAC Address

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THEORY

1. What is Ethernet?

It is a widely used technology used in local area networks (wired) , it defines how devices communicate over ethernet network.

2. What is an Ethernet frame?

It contains:

Destination: layer 2 (mac) address to which the frame is being sent.

Source: layer 2 (mac) address of the device which sent the frame.

Type/length: it indicates layer 3 protocol used in the encapsulated packet(like ipv4 or ipv6).

Payload: the actual data.

FCS: Frame Check Sequence: used to test if an error happened while transferring the packet (runs a CRC algorithm).

3. What is a MAC address?

Layer 2 address used in LANs

4. What does MAC stand for?

Media Access Control.

5. Difference between a MAC address and an IP address?

Mac address: it is a physical address of layer 2 and it is assigned to the device when it is made,used in local interface identification.

IP address: it is a logical address of layer 3,used in logical addressing and routing.

6. What is a MAC Address Table?

It adds a mac address and associate it to port of a switch.

7. How does a switch learn MAC addresses?

A pc sends a frame containing the source mac address to a specific port of the switch then the switch map this mac address with the port in the mac address table.

8. What is a source MAC address?

It is the mac address of the device that sent the frame.

9. What is a destination MAC address?

It is the mac address of the device to which the frame is sent to.

10. What is an unknown unicast frame?

When the switch doesn't have an entry in its mac address table.

11. What is flooding?

When an unknown unicast frame is sent the switch has to send it to multiple ports except the one the frame was received on from and this is called **flooding**.

12. What is known unicast forwarding?

It is when the destination mac address is already present in a mac address table of the switch.

13. At which OSI layer does Ethernet switching primarily operate?

Layer 2 — data link layer.

14. Why does a switch maintain a MAC Address Table?

It helps the switch determine where frames should be forwarded.

15. Why is Ethernet knowledge important in cybersecurity?

Security professionals learn to examine MAC addresses, Ethernet frames, local network traffic, ARP tables, switch behaviour, and packet captures. They also learn about network

segmentation and suspicious Layer 2 activity. These skills are useful later for Wireshark, Nmap, incident response, vulnerability assessment, and penetration testing.

PRACTICAL IDENTIFICATION

Record: Network adapter name, MAC address, IPv4 address, default gateway.

Network Adapter name: Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC.

Mac address: C4-13-75-3C-F1-01

IPv4 address: 192.168.100.209

Default gateway: 192.168.100.1

SCREENSHOTS

Attach screenshots of: getmac, ipconfig /all, arp -a

1.getmac

```
PS C:\Windows\System32> getmac

Physical Address    Transport Name
=====
C4-13-75-3C-F1-01   \Device\Tcpip_{4FE709A8-2353-40ED-82BF-62D544AC18C8}
C4-13-75-3C-F1-02   Media disconnected
00-FF-74-4B-36-3C   Media disconnected
0A-00-27-00-00-16   \Device\Tcpip_{F52C12F9-F595-42A9-A285-7416B958E8E6}
```

2.ipconfig /all

```
PS C:\Windows\System32> ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-AM22QKJ6
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-16
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::dd09:f6fe:ee7f:b4ac%22(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 789184551
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
NetBIOS over Tcpip. . . . . : Enabled
```

Unknown adapter McAfee VPN:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-74-4B-36-3C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 4:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #4
Physical Address. . . . . : C6-13-75-3C-F1-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 5:

```
Media State . . . . . : Media disconnected
```

Wireless LAN adapter Wi-Fi 5:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC #5
Physical Address. : CA-13-75-3C-F1-01
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Realtek 8851BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. : C4-13-75-3C-F1-01
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::e3bb:c8ce:a438:c2e7%5(Preferred)
IPv4 Address. : 192.168.100.209(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : Thursday, September 10, 2026 6:48:38 PM
Lease Expires : Saturday, September 12, 2026 3:20:24 PM
Default Gateway : 192.168.100.1
DHCP Server : 192.168.100.1
DHCPv6 IAID : 113513333
DHCPv6 Client DUID. : 00-01-00-01-31-55-B1-2C-00-FD-6B-09-4A-82
DNS Servers : fe80::1%5
 8.8.8.8
 8.8.4.4
 fe80::f912:e97b:33cc:e623%5
 fe80::1%5
NetBIOS over Tcpip. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : C4-13-75-3C-F1-02
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

3.arp -a

```
PS C:\Windows\System32> arp -a
```

```
Interface: 192.168.100.209 --- 0x5
```

Internet Address	Physical Address	Type
192.168.1.250	c0-fd-84-cc-e6-23	dynamic
192.168.100.1	60-9b-b4-7a-74-7a	dynamic
192.168.100.37	34-2e-b7-94-cc-3f	dynamic
192.168.100.84	e0-03-6b-61-61-fe	dynamic
192.168.100.255	ff-ff-ff-ff-ff-ff	static
224.0.0.7	01-00-5e-00-00-07	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
Interface: 192.168.56.1 --- 0x16
```

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.7	01-00-5e-00-00-07	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

```
PS C:\Windows\System32> |
```